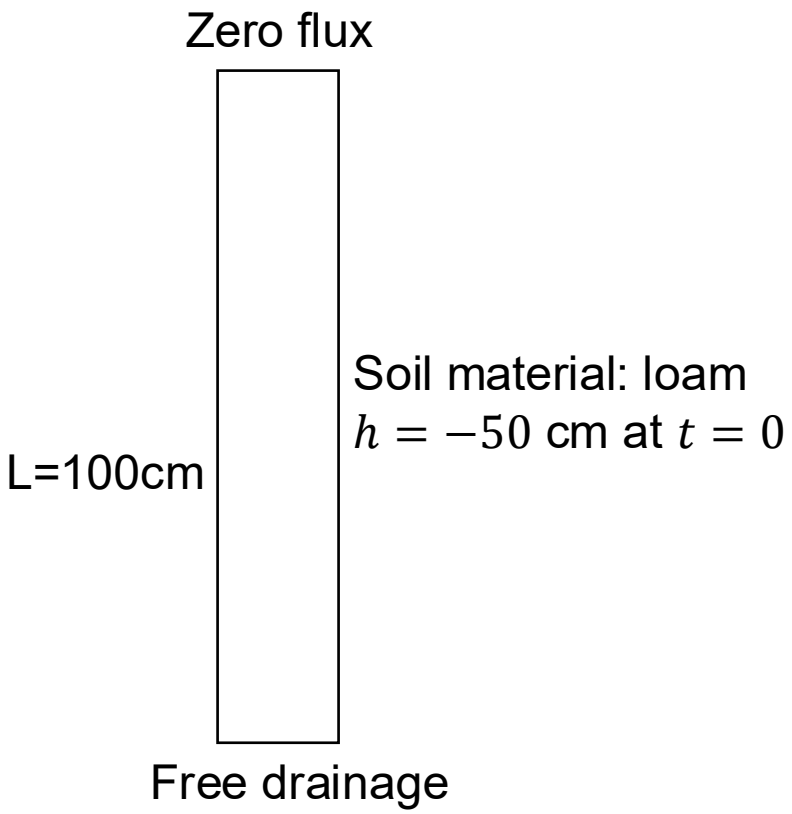
HWRS 561b: Physical Hydrogeology II

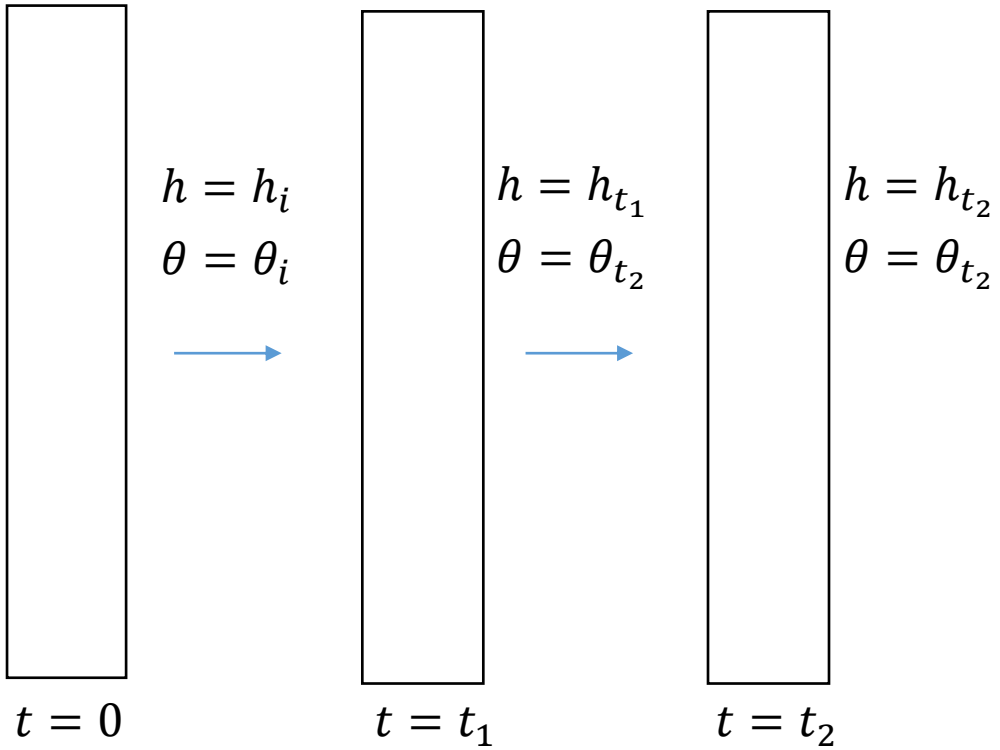
Multidimensional unsat flow

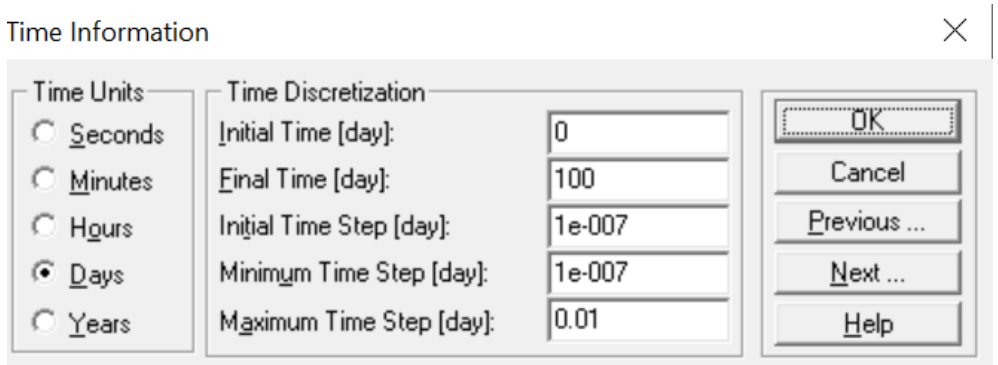
Agenda: 2D transient unsaturated flow
Reading: Jury & Horton, Chapter 4

Delayed Drainage

Question: Is the "unit gradient" a reasonable assumption during drainage?





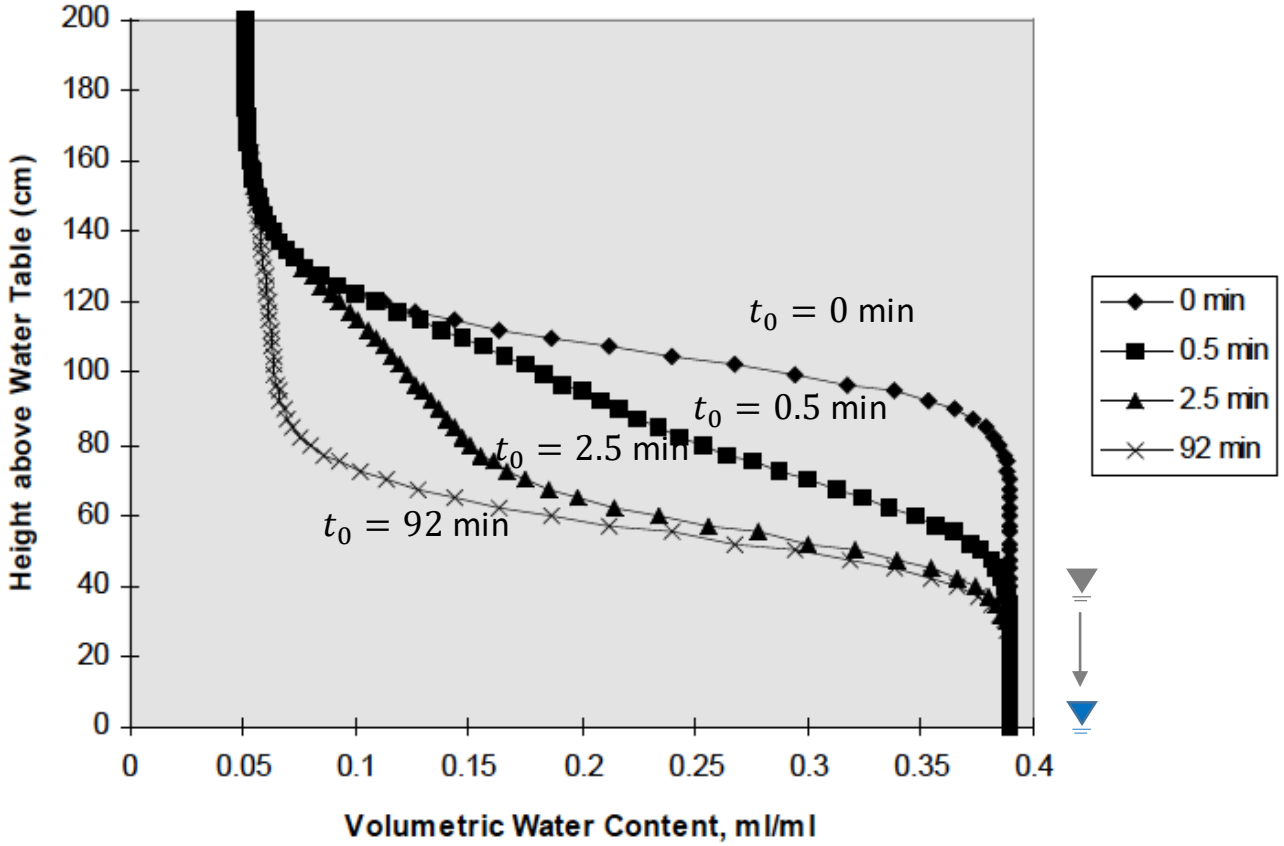


Note: "unit gradient" means flow is only driven by gravity

$$q = -K\left(\cancelto{0}{\frac{\partial h}{\partial z}} + 1\right) = -K$$

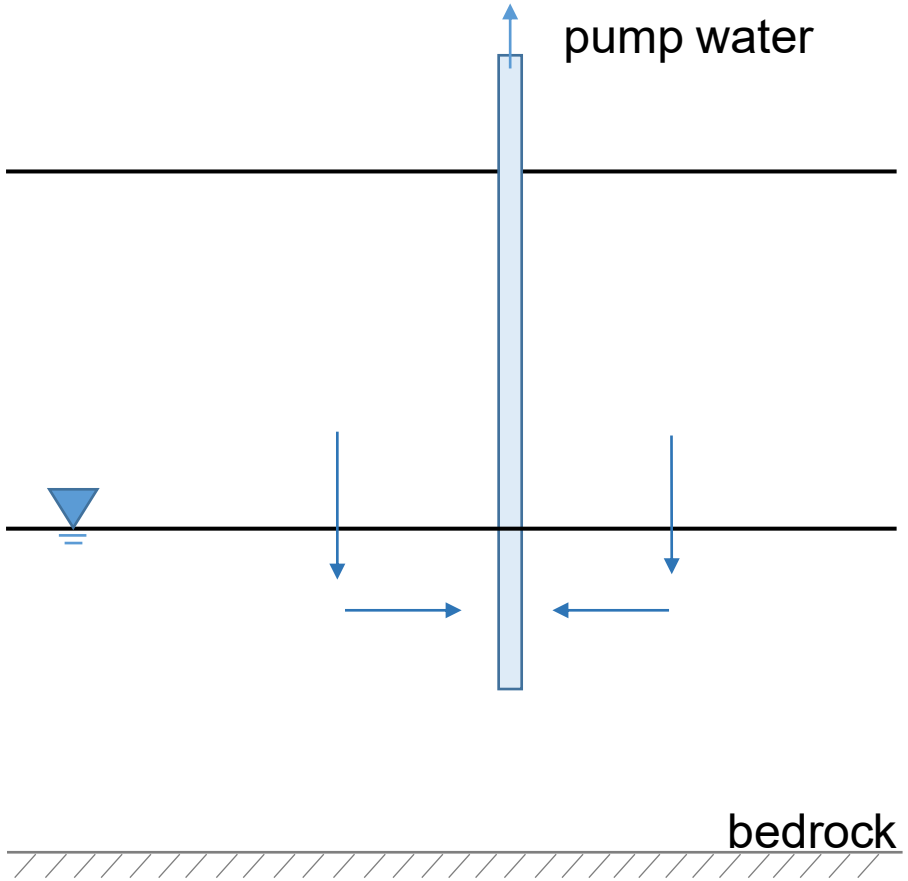
Delayed Drainage

1-D delayed drainage modeling for Borden aquifer parameters

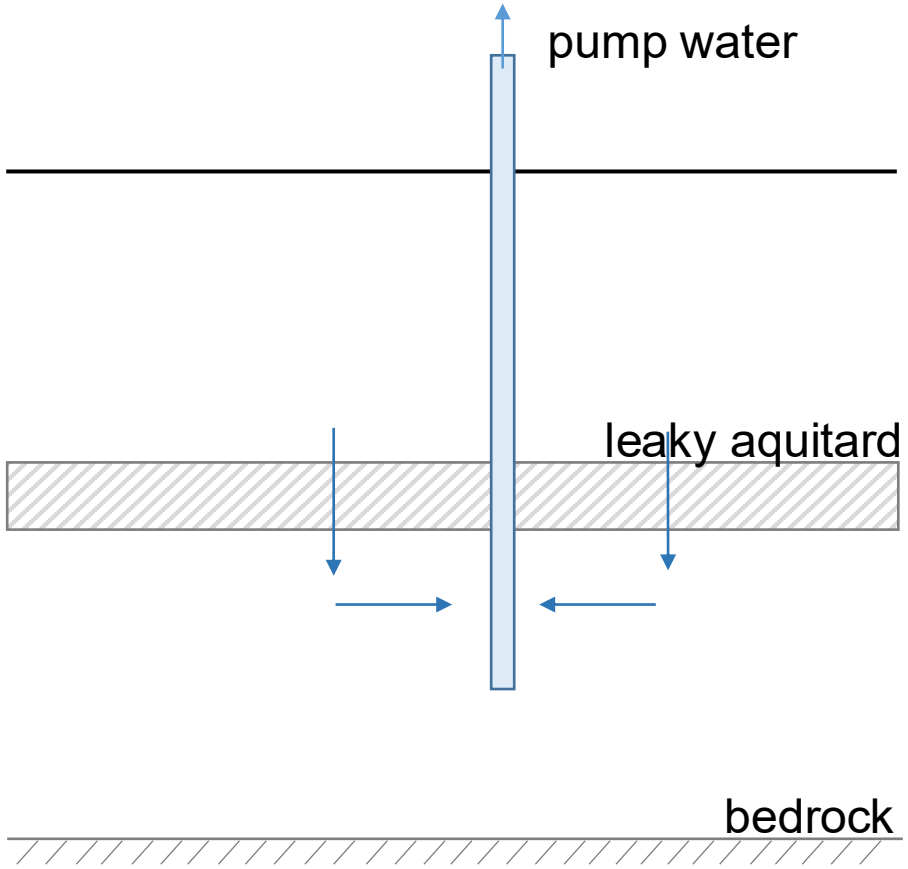


There is a time scale involved for unsaturated water to respond to change in water table

Pumping Tests: Unconfined Aquifers

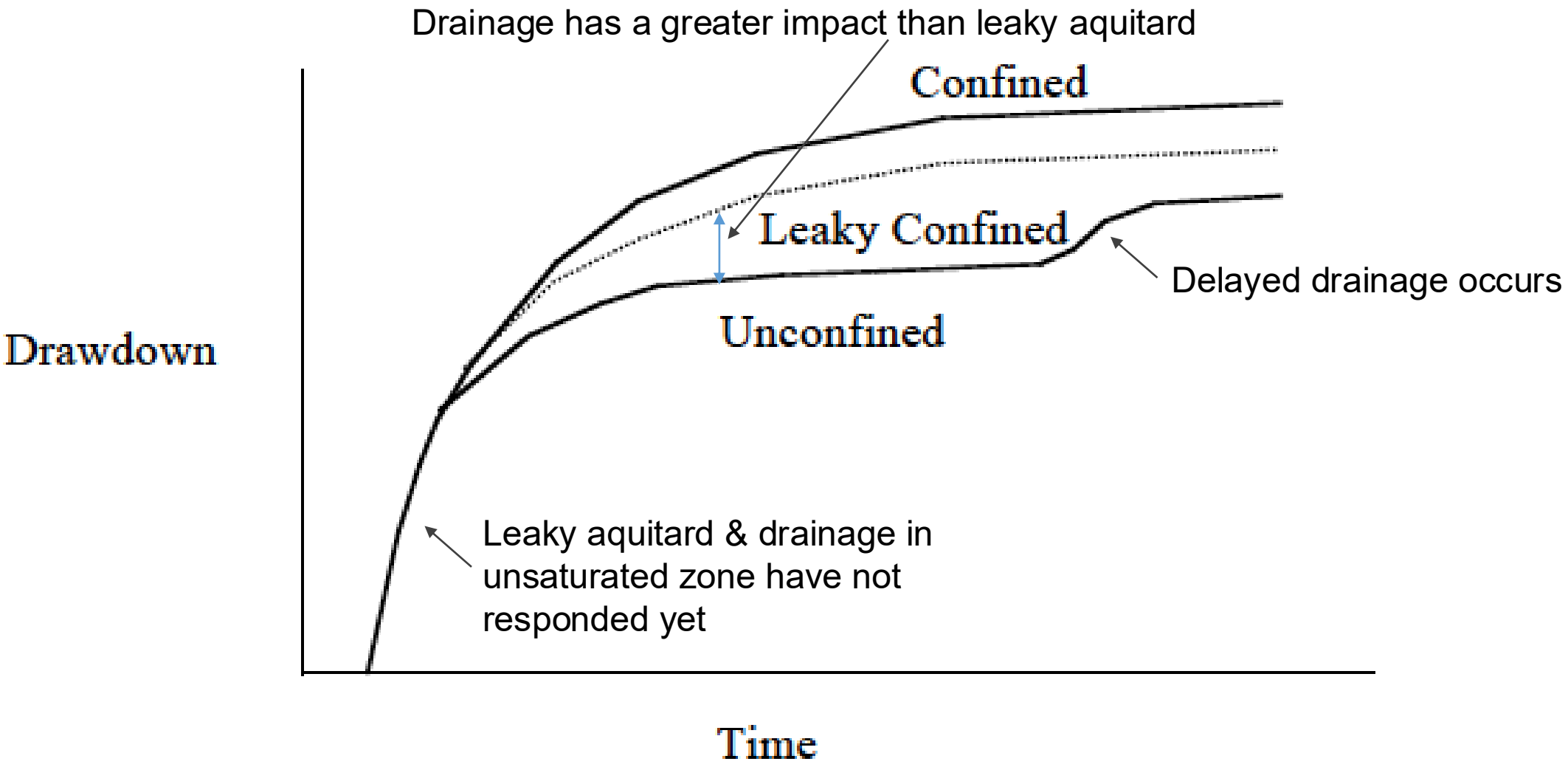


Unconfined aquifer

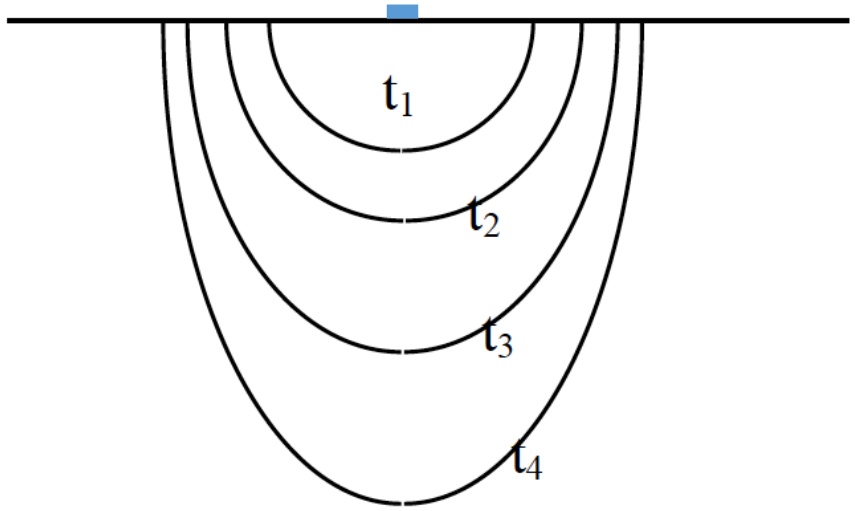


Confined aquifer w/ leaky aquitard

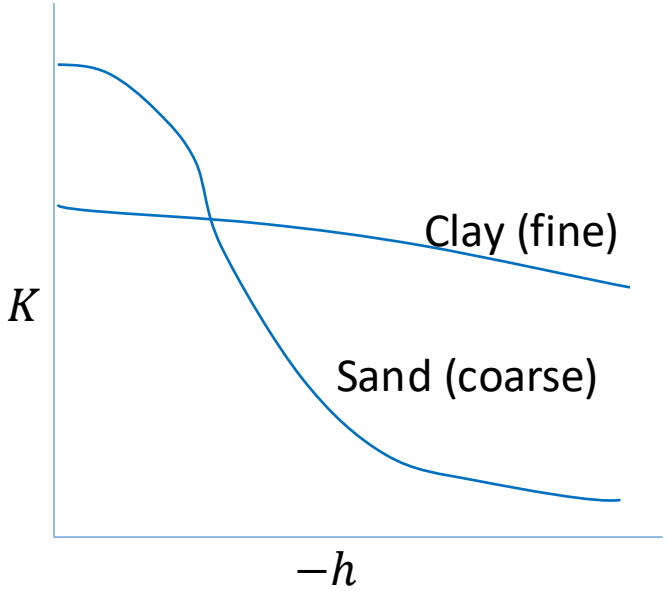
Pumping Tests: Unconfined Aquifers



Infiltration from a Point Source



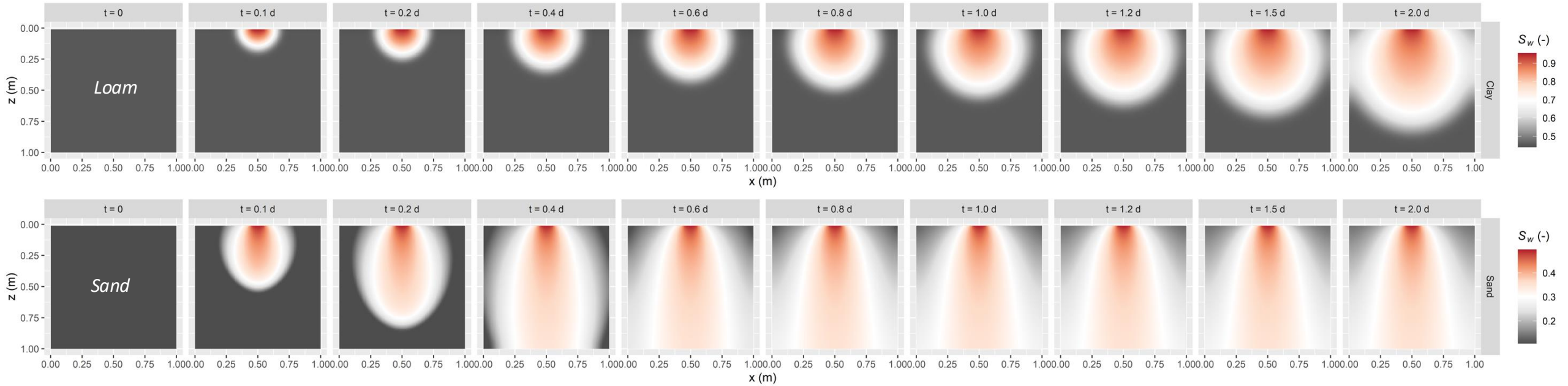
Horizontal flow: driven by capillary pressure
Vertical flow: driven by capillary pressure and gravity



***Question**:* How will the wetted "bulb" be different for sand and clay?

- Capillary pressure driven flow is stronger in clay

S_w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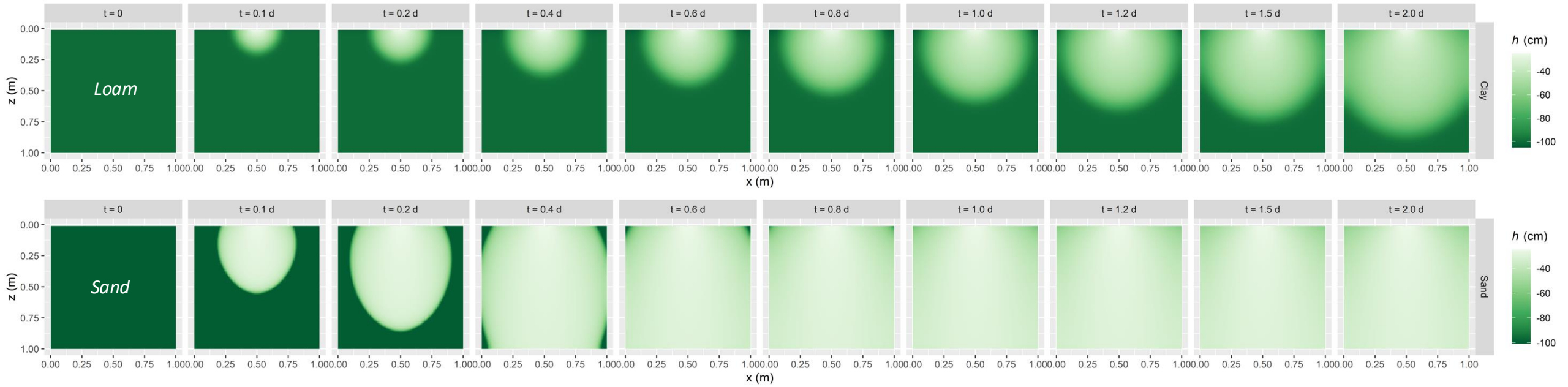
t = 0.1 d
t = 0.2 d
t = 0.4 d
t = 0.6 d
t = 0.8 d
t = 1.0 d
t = 1.2 d
t = 1.5 d
t = 2.0 d
Loam
Sand
z (m)
x (m)
Clay
Sand
S_w (-)

h

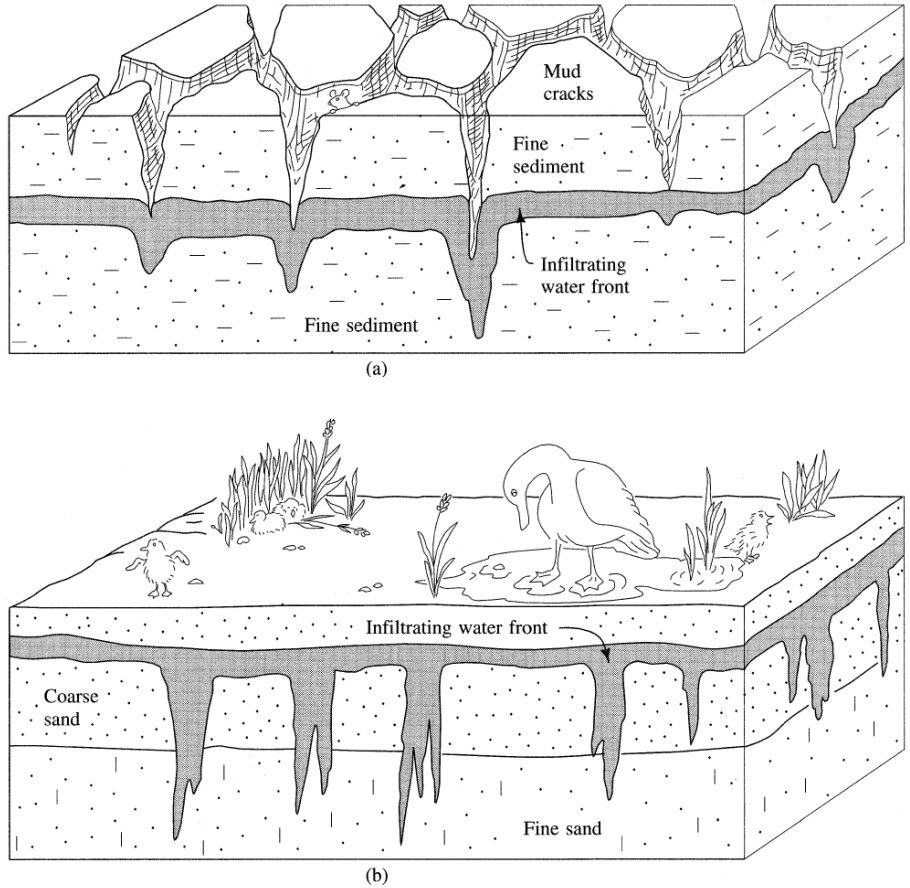

t = 0
t = 0.1 d
t = 0.2 d
t = 0.4 d
t = 0.6 d
t = 0.8 d
t = 1.0 d
t = 1.2 d
t = 1.5 d
t = 2.0 d
Loam
Sand
z (m)
x (m)
Clay
Sand
h (cm)

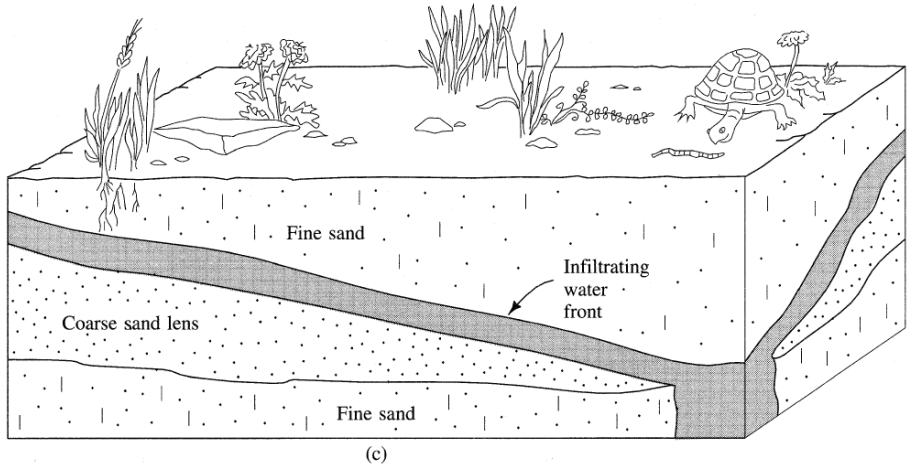
Preferential flow

Preferential flow: Significantly greater than average downward movement of water through part of the soil during an infiltration or drainage event.

Three types of preferential flow: Macropore flow, funnel flow, and unstable flow.

FIGURE 4.18 Preferential water movement in the vadose zone due to (a) short circuiting, (b) fingering, and (c) funneling.





Macropore flow

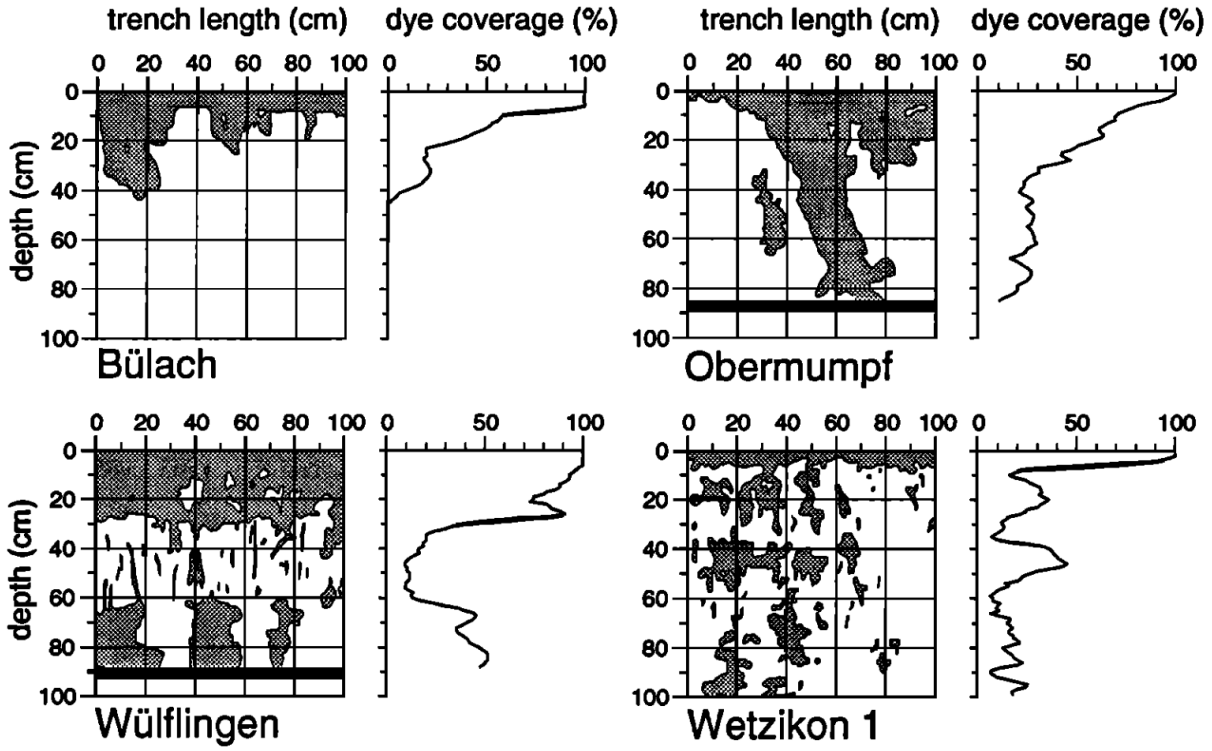


Figure 2. Vertical flow patterns of Brilliant Blue FCF and one-dimensional profiles of dye coverage after sprinkling application of 40-mm colored water in four different soils. These examples represent "wet" initial conditions. The solid bar at Obermumpf and Wülflingen indicates the maximum depth of excavation.

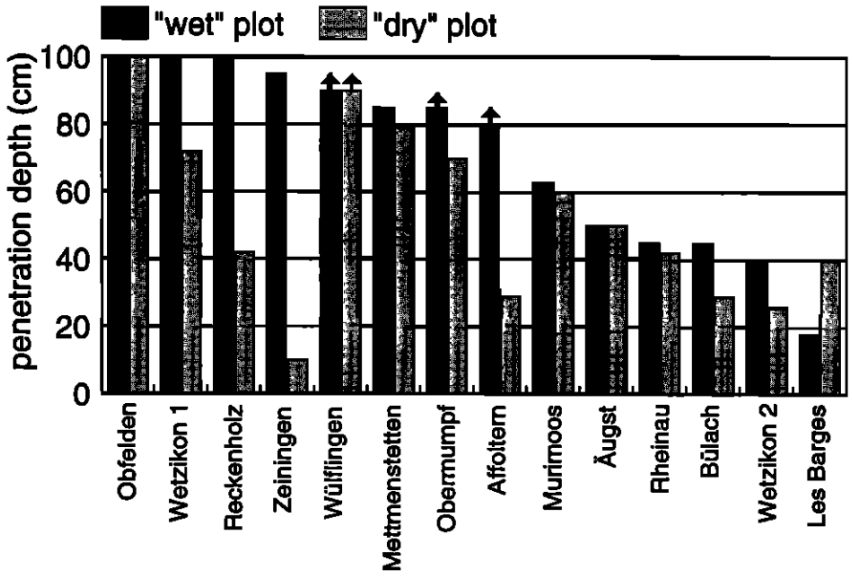
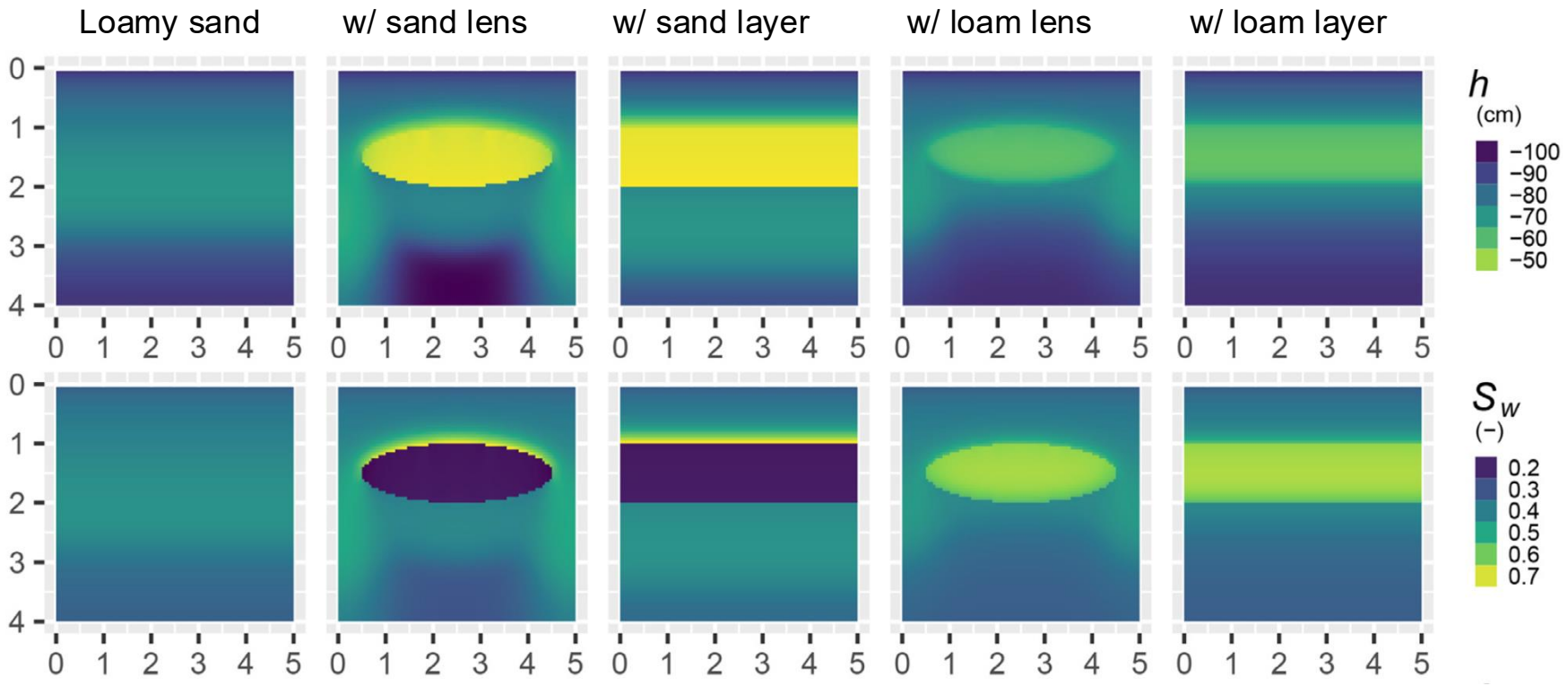


Figure 5. Maximum depth of percolation of Brilliant Blue FCF in the different soils. The sites are ranked according to the percolation depth of the "wet" plot. Big stones or erratic boulders hindered deeper excavation at the Wülflingen, Obermumpf, and Affoltern sites. However, it was expected that the dye had moved deeper than the maximum excavation depth at these sites, as indicated by the arrows.

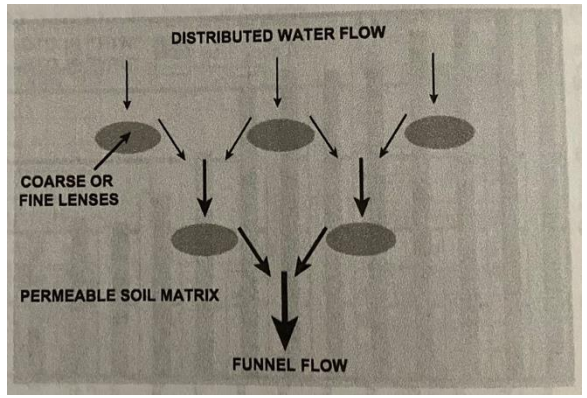
Flury et al., 1994

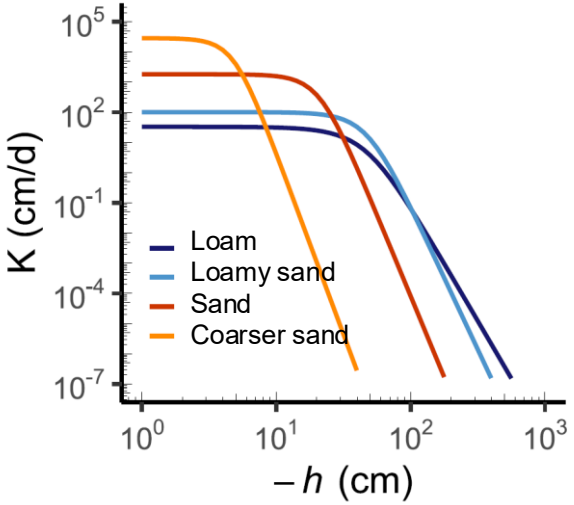
- 4 cm of water infiltration
- Channeling at various degrees observed for structured soils (moderate to extreme)
- For a uniform water content $\theta = 0.25$. Penetration should be 16 cm for 4 cm of water infiltration

Funnel flow



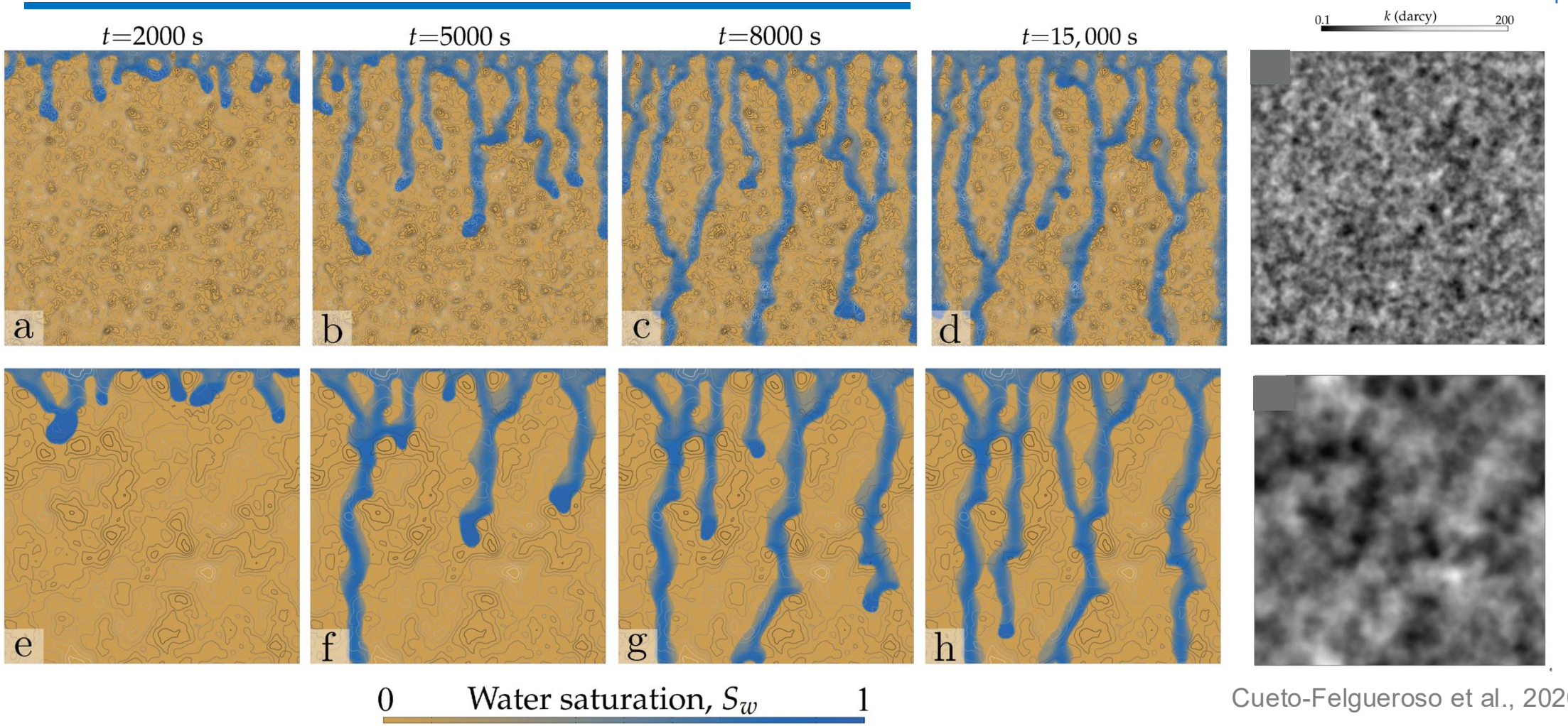
Zeng & Guo, 2021





- Both sand and loam act as a barrier for water flow in loamy sand, but the mechanisms are different.
- Sand acting as a barrier generates preferential flow on the side (the so-called “funnel flow”; Kung (1990)). See Chapter 4 of Jury & Horton.

Unstable flow



Cueto-Felgueroso et al., 2020

- Unstable flow is distinct from other forms of preferential flow. It is a flow phenomenon whose extreme flow location is not a consequence of permeability variations in the porous medium.